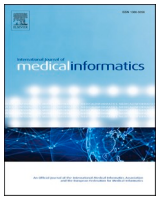
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Large language models can support generation of standardized discharge summaries – A retrospective study utilizing ChatGPT-4 and electronic health records

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ABSTRACT

Objective: To evaluate whether psychiatric discharge summaries (DS) generated with ChatGPT-4 from electronic health records (EHR) can match the quality of DS written by psychiatric residents.
Methods: At a psychiatric primary care hospital, we compared 20 inpatient DS, written by residents, to those written with ChatGPT-4 from pseudonymized residents' notes of the patients' EHRs and a standardized prompt. 8 blinded psychiatry specialists rated both versions on a custom Likert scale from 1 to 5 across 15 quality subcategories. The primary outcome was the overall rating difference between the two groups. The secondary outcomes were the rating differences at the level of individual question, case, and rater.
Results: Human-written DS were rated significantly higher than AI (mean ratings: human 3.78, AI 3.12, $p < 0.05$). They surpassed AI significantly in 12/15 questions and 16/20 cases and were favored significantly by 7/8 raters. For "low expected correction effort", human DS were rated as 67 % favorable, 19 % neutral, and 14 % unfavorable, whereas AI-DS were rated as 22 % favorable, 33 % neutral, and 45 % unfavorable. Hallucinations were present in 40 % of AI-DS, with 37.5 % deemed highly clinically relevant. Minor content mistakes were found in 30 % of AI and 10 % of human DS. Raters correctly identified AI-DS with 81 % sensitivity and 75 % specificity.
Discussion: Overall, AI-DS did not match the quality of resident-written DS but performed similarly in 20% of cases and were rated as favorable for "low expected correction effort" in 22% of cases. AI-DS lacked most in content specificity, ability to distill key case information, and coherence but performed adequately in conciseness, adherence to formalities, relevance of included content, and form.
Conclusion: LLM-written DS show potential as templates for physicians to finalize, potentially saving time in the future.

1. Introduction

Documentation burden for healthcare workers has continued to increase in recent years [1,2]. It currently consumes a substantial portion of physicians' working hours [3] and has been linked to overwork, emotional exhaustion, and burnout [4–6]. As physicians spend considerable time writing discharge summaries (DS) [7,8], with one study measuring a mean time of 18.25 min for somatic medical DS [7], it is worth exploring whether large language models (LLMs) can support physicians in this task to reduce workload.

The capabilities of LLMs in text-based tasks have improved significantly in recent years. The trajectory is impressive: For instance, in a 2023 experiment, OpenAI's ChatGPT-4 correctly answered > 86 % of questions from a test material sample of the USMLE (United States Medical Licensing Examination), whereas GPT-3.5 had performed close to the passing threshold of 60 % just one year prior [9,10]. Recently,

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LLMs have been applied to various medical tasks, such as differential diagnosis [11], triage [12,13], literature search [14], documentation [15–17], and DS [18–20]. Several studies have shown that the quality of DS can be improved through standardization [21–23], making them a suitable target for LLMs.

In a study from 2022, Cait et al.[24] compared 7 different automatic summarization models for the generation of brief, patient-directed hospital after-visit summaries. Clinical ratings by 2 medical experts of the top-performing LED (Longformer-Encoder-Decoder) model revealed hallucinations in 28 % out of 18 cases, but high scores for expected ease of revision in > 75 % of cases, suggesting potential to reduce physicians' workload.

In a 2023 study, Hartman et al.[19] employed BART and BERT-transformers to generate 25 neurological DS from electronic health records (EHRs). The models were fine-tuned on a custom dataset from 6,600 hospital admissions. They were evaluated by 2 blinded board-certified specialists who rated both DS versions on a Likert scale for overall quality, readability, factuality, and completeness. 62 % of LLM-generated DS met the standard of care, defined as scores ≥ 7/10 in all categories, compared to 94 % of physician-written DS. However, generation of LLM-DS was directly dependent on previously written physician-DS, making a direct comparison difficult.

Another study by Williams et al.[18] in 2024 compared 100 DS written with ChatGPT-4 to ChatGPT-3.5-turbo based on EHRs < 3.500 tokens from 100 emergency department visits, using a brief instructional prompt. There was no comparison to physician-written DS. For the evaluation, 2 unblinded resident physicians determined occurrence of inaccuracies, hallucinations, and omission of relevant clinical information. In this study, 33 % of ChatGPT-4 and 10 % of ChatGPT-3.5-turbo-written DS were found to be completely free of all the three predefined deficiencies. 42 % of ChatGPT-4-DS and 64 % of ChatGPT-3.5-written DS exhibited hallucinations.

In 2024, Tung et al.[20] compared 5 DS written with ChatGPT-4 and a detailed prompt to those written by junior physicians in a urology clinic, based on fictional EHR notes. The EHRs were written in bullet points and ranged from 424 to 1,110 tokens. 5 blinded primary care physicians rated the DS across 6 categories. ChatGPT-4 performed significantly better in information provision, with no significant differences in clarity, collegiality, conciseness, follow-up recommendations, or overall satisfaction. No instances of hallucinations were observed.

1.1. Objective

Our objective was to determine whether DS, generated with ChatGPT-4 based on psychiatric residents' EHR notes and a standardized prompt, could match the quality of DS written by the same residents. It is unknown how LLM-generated DS perform in a psychiatric clinical setting, which poses specific challenges such as detailed psychopathological descriptions, text similarities in notes of repeated standardized risk stratification, and often lengthy EHRs.

In reference to the previously mentioned studies [18–20,24], we aimed to directly compare LLM and resident-written DS, utilize pseudonymized real-life EHRs, develop a custom prompt tailored to the specific requirements of psychiatric DS and our clinic's standards, and use a detailed Likert scale to better assess the specific strengths and weaknesses of LLM-DS.

Our 5-point Likert scale consisted of 15 items, each measuring subcategories of overall DS quality. An additional question determined raters' ability to correctly differentiate between ChatGPT-4 and resident-DS.

To increase rater neutrality, we chose a two-step evaluation process: In the first step, 8 blinded psychiatry specialists with no prior case knowledge rated 20 resident-written against 20 ChatGPT-4-written DS on the Likert scale. The blinding and high number of raters were chosen to increase objectivity and reduce the impact of individual rater bias. However, having no detailed case knowledge would negatively impact raters' ability to judge whether all case-relevant information had been included in the DS. To mitigate this effect, raters were provided with "clinical assessments" for each case (see 2.4 Study Design). In the second evaluation step, the investigators conducted an unblinded case-based review, comparing both DS versions qualitatively with their respective EHRs and screening for hallucinations and other content errors.

For the null hypothesis, we posited that there would be no significant overall rating difference between the two groups. The primary outcome was the difference in median ranks across all questions between both groups, as determined by a Wilcoxon signed-rank test. The secondary outcomes were the differences in median ranks between both groups at the level of individual rating question, case, and rater. From the available LLMs, we chose ChatGPT-4 because of its high accessibility and overall performance in language-dependent tasks, particularly content comprehension and general knowledge, as has been demonstrated in benchmarks like DROP and MMLU [25].

2. Methods

2.1. Definitions

In this study, the psychiatric residents' DS group was labeled "*Human*" and the ChatGPT-4 DS group was labeled "*AI*".

DS were defined as comprehensive summaries of the inpatient's treatment and were required to consist of three segments to align with our clinic's standards:

1. A description of all individual diagnoses with their respective treatments.
2. An additional short overall case summarization.
3. Further recommendations.

"Discharge *letters*" (see 2.3 Study Design) were defined as the complete discharge documents comprising two parts:

1. The "clinical assessment" section (including current complaints, patient history, physical, psychopathological, and instrumental diagnostic findings) and
2. the DS.

Hallucinations were defined as untrue statements caused by either critical misinterpretation of existing or fabrication of new information.

2.2. Study population

We retrospectively analyzed 30 EHRs from adult inpatients, age 18 to 65, who had been treated on an acute psychiatric ward at the Psychiatric University Hospital Zurich in 2023. All psychiatric main diagnoses were eligible for inclusion. To provide a sufficiently informative foundation for ChatGPT-4, we only included cases > 1,000 tokens and in which psychiatric residents' EHR notes were deemed adequately informative, based on a subjective review by the principal investigator. Cases with EHRs > 5,000 tokens were excluded. This token range was set to account for the maximum context size of 8,192 tokens in GPT-4–0613, a prompt around 1,000 tokens, and space for up to 2,000 tokens for the DS. We screened a total of 30 cases for inclusion, randomly selected 10 cases to develop the custom prompt, and then used the remaining 20 cases for the per-protocol analysis. The cases with their diagnoses and input token counts are shown in Supplementary Tables 1 and 2.

2.3. Study design

We based the 15 questions of our Likert scale on earlier studies that had investigated the usefulness of standardized DS, as well as our clinic's internal DS guidelines [26–28]. Each question was scored between 1 and 5 with the following definitions:

1 = I strongly disagree,
2 = I mostly disagree,
3 = I neither agree nor disagree (neutral),
4 = I mostly agree,
5 = I strongly agree.

We prespecified two reference questions that we expected to be the most indicative of overall DS quality: "DS ready to send" (Q15) as the primary indicator, and "low expected correction effort" (Q14) as the secondary indicator. The questions are depicted in Table 1.

To develop the custom prompt for ChatGPT-4, we used a training sample of 10 randomly selected cases, incorporating insights from prior studies that had investigated content and structuring of standardized DS [22,26,29], as well as our clinic's internal DS guidelines. The cases covered a diverse range of diagnoses and input token counts. The custom prompt was refined over 50 generations during the training period, based on the principal investigator's subjective evaluation of DS outputs. The final prompt comprised 974 tokens and is detailed in Table 2. It contained instructions on structure, diagnostic criteria, content inclusion/exclusion, timelines, and tone, among other aspects. After prompt finalization, *AI*-DS completely adhered to the required structural format and therefore, this aspect was not included in the formal rating.

For the evaluation sample, *AI*-DS were generated from the residents' pseudonymized EHR notes and the standardized prompt with ChatGPT-4. For all DS generated, we used the same GPT-4–0613 build between the November 21, 2023, and January 10, 2024, updates.

To account for the temperature setting, the best out of three outputs was used for each case, based on the subjective judgment by the principal investigator. To provide all raters with background information for the cases, they were always presented with complete *Human* and *AI* discharge *letters* containing the identical "clinical assessment" section. The *Human* discharge letters were only pseudonymized, while for the *AI* discharge letters, the "clinical assessment" from the *Human* letters was added to the *AI* DS. The resulting two letters only differed in their DS section, and their complete workflows are depicted in Fig. 1. Raters then only evaluated DS, as this segment typically represents the primary workload for physicians in routine practice, with the "clinical assessment" often being directly derived from other documents.

Table 1
Questions Q1–15 rated on the Likert scale to assess sub-categories of DS quality and question Q16 to assess raters' ability to correctly identify *AI* vs. *Human* for each case.

Question Number	Description
Q1	The text describing each diagnosis is assigned correctly (it does not fit better to a different diagnosis).
Q2	The relevant clinical findings are mentioned.
Q3	The relevant instrumental and laboratory findings are mentioned.
Q4	The relevant therapies that were carried out are mentioned.
Q5	The relevant therapeutic effects and side effects are mentioned.
Q6	The summary gives me a good overview of the most important information of this case.
Q7	The relevant recommendations for further psychiatric treatment are mentioned, if possible, with the name of outpatient practitioner and an appointment.
Q8	The relevant recommendations for further non-psychiatric treatment are mentioned, if possible, with the name of outpatient practitioner and an appointment.
Q9	The text contains an appropriate amount of professional medical terminology.
Q10	Relevant safety concerns (endangering oneself or others) and other legally relevant information are mentioned.
Q11	The text is coherent.
Q12	Information is formulated specifically (and not too generally).
Q13	The text contains little unnecessary information.
Q14	As a senior doctor, I would only have to make few corrections.
Q15	I would send out this discharge summary in its current form.
Q16	I am sure that this currently rated version was written by AI.

2.4. Ethical considerations and data protection

All patients provided written general consent for the use of their healthcare data for research purposes. To protect the privacy of patients and employees, all EHR notes used in ChatGPT-4, and all original physician-written discharge letters were manually pseudonymized, and non-relevant information was removed in accordance with the guidelines of the national ethics committee and the Swiss Personalized Health Network [30,31]. The study was approved by the ethics committee of the Canton of Zurich (BASEC-ID 2024–00371).

2.5. Statistical methods

We calculated standardized Cronbach's α for internal consistency of the Likert scale and used standard deviations and intraclass correlation coefficients (ICCs) to assess inter-rater agreement across main groups and secondary groups (question, case, and rater levels).

To assess relationships between diagnoses, EHR lengths, and ratings, we first determined that only the two most common diagnoses – F20.0 and F31.2 (5 and 4 cases) – would be suitable for analysis as all other cases (9 individual diagnoses, 2 with F20.2) would be too prone to bias due to the small sample size.

Shapiro-Wilk tests showed normal distribution for EHR lengths within both, F20.0 and F31.2 cases ($W_{F20.0} = 0.92$, $p = 0.54$; $W_{F31.2} = 0.92$, $p = 0.51$), as well as for their ratings ($p_{F20.0AI} = 0.16$; $p_{F31.2AI} = 0.66$; $p_{F20.0Human} = 0.77$; $p_{F31.2Human} = 0.34$). We then performed t-tests to assess differences in EHR lengths and ratings between the F20.0 and F31.2 groups. Lastly, we calculated Spearman correlation between the EHR lengths of all 20 cases and their ratings to assess their dependence.

For the primary outcome comparing the overall ratings of *Human* and *AI*, and for the secondary outcomes, we again assessed normality, which was rejected in both main and all subgroups (all with $p < 0.001$).

For the primary outcome, we consequently used a Wilcoxon signed-rank test to assess the median rank differences between both groups, with p set at < 0.05 for significant rating differences.

For the secondary outcomes, we applied Wilcoxon signed-rank tests at the level of each question, case, and rater. Given the greater number of variables, we here applied a Bonferroni adjustment with lower p-values to reduce the risk of Type 1 errors, with $p = 0.0033$ (0.05/15) in the per-question, $p = 0.0025$ (0.05/20) in the per-case, and $p = 0.00625$ (0.05/8) in the per-rater analysis.

For the two most important questions, Q15 "DS ready to send" and Q14 "low expected correction effort" specifically, we additionally calculated overall approval ratings by combining ratings ≥ 4 as "favorable" and ≤ 2 as "unfavorable". Finally, we calculated sensitivity and specificity for correct *AI* identification, based on raters' confidence ratings (Q16), using ratings ≥ 4 for *AI* and ≤ 2 for *Human* as thresholds.

3. Results

3.1. Analysis of inter-rater agreement, rating scale, and relationships between diagnoses, EHR lengths, and ratings

For overall ratings, the ICCs between raters in both main groups were low (0.25 for *Human*; 0.23 for *AI*). For the individual questions, ICCs ranged from medium to high in *Human* (0.48–0.86) and medium to low in *AI* (−0.02–0.59). The ICCs per question rating are depicted in Table 4. The standardized Cronbach's α was 0.91, indicating high internal consistency of the rating scale. The item statistics for all individual questions are depicted in the Supplementary Table 3. Standard deviations (SD) did not differ significantly between the two groups across the entire sample ($SD_{Human} = 1.05$, $SD_{AI} = 0.99$, $p = 0.009$) and showed comparable ranges at the levels of individual question, case, and rater (see Table 3).

There was no significant difference in EHR lengths between the two most common diagnosis groups, F20.0 and F31.2 ($p = 0.58$; $mean_{F20.0} =$

Table 2
The custom prompt developed for ChatGPT-4 to generate the DS.

Your Role: You are a resident physician tasked with writing a discharge summary of a patient from a psychiatric clinic from the patient's electronic health records. The diagnoses are:
Diagnosis 1,
Diagnosis 2,
Diagnosis 3

Write a discharge summary from the following notes as a continuous text in complete sentences. Write only the last section "Procedures" in bullet points. All sections other than "Procedures" should not contain bullet points.

Form: Start with the treatment period. Structure the medical summary into 3 sections: Diagnoses with corresponding ICD-10 codes, assessment, Procedures. Write the text for each diagnosis in a block of text without structural subdivisions.

Diagnoses: Write a short explanatory text under each diagnosis, addressing the following points: Description of the syndrome with relevant psychopathological findings, instrumental findings, and laboratory results, then detailed current medication therapy with justification for indication, precise dosage, observed effects, and side effects, if reported. Mention the start of each medication, changes of dosage, or changes of medication with the date and reason for decision, if apparent from the notes. Write medications as active ingredients and not as brand names. If isolation measures were implemented, mention this. Then explain when and how it was justified. Then write whether the diagnostic criteria according to ICD-10 are fully met. If they are not fully met, write how many of the necessary criteria are fulfilled.

Summary: Then under "Summary", write a summarization of the entire treatment. Explicitly state whether, at the time of discharge, there were any acute indications for self-harm or harm to others, particularly in terms of suicidality. Only if the discharge occurred through absconding, state this fact explicitly.

Procedures: Then under "Procedures," write recommendations for further treatment, names of further caregivers, and timing of further treatment, if mentioned in the progress notes.

Style: Write the text in the style of a discharge letter using medical terminology. Proceed chronologically, write factually, objectively and concretely, formally, professionally, politely, respectfully. Avoid generalizations, assumptions, verbose and cumbersome formulations, and redundancy. Do not invent or presume anything. Generally, formulate in the imperfect tense, reported speech from individuals (e.g., from patients) in subjunctive I, and medical assessments and diagnoses in the present tense.

Further instructions: Only present content, interpretations, assumptions, and recommendations that directly emerge from the physicians' notes; do not generate recommendations on your own or add anything. Write nothing about topics for which there is no information basis from the notes.

Explanation of abbreviations: BE = Blood Extraction, DUP = Drug Urine Sample, BEP = Stress Test, TU = Day Leave, SA = Social Worker, PP = Nursing Staff, HA = General Practitioner, AA/AÄ = Assistant Doctor, OAV = Senior Doctor Rounds, OA/OÄ = Senior Doctor

The EHR notes follow now:

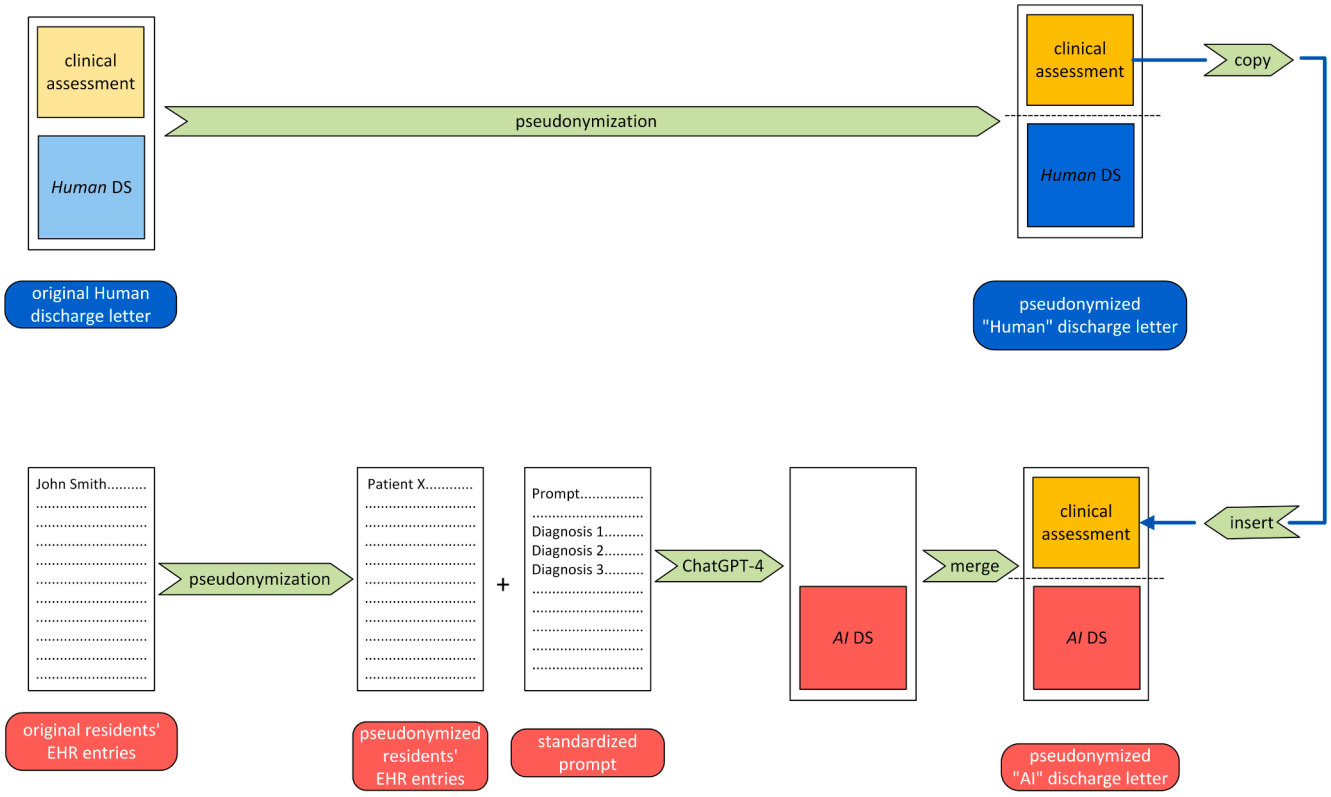


Fig. 1. The upper section depicts the workflow for generating the pseudonymized *Human* discharge letters. The lower section illustrates the workflow for generating the pseudonymized *AI* discharge summaries and letters.

Table 3
Standard deviations (SD) for *Human* and *AI* of the entire sample (Total) and SD ranges per individual question, case and rater.

Study Group	SD Total	SD per Question	SD per Case	SD per Rater
Human	1.05	0.69–1.19	0.94–1.26	0.63–1.37
AI	0.99	0.73–1.13	0.94–1.18	0.76–1.39

2,795 tokens; $mean_{F31.2}$ = 3,256.5 tokens).

There was also no significant difference in the ratings of F20.0 and F31.2 cases, for both *Human* and *AI* (p_{Human} = 0.73; p_{AI} = 0.51). Across all 20 individual cases, no relevant Spearman correlation was found between their EHR lengths and ratings in either *Human* or *AI* (rho_{Human} = 0.05; rho_{AI} = 0.04). The relationships among these three variables are visualized in Supplementary Fig. 1.

Raters correctly identified *AI* with 81 % sensitivity and 75 % specificity (AUC 0.92).

3.2. Main group analysis

The main group analysis showed significant overall rating differences between *Human* and *AI* in favor of *Human* ($p < 0.05$), rejecting the null hypothesis. Mean ratings were 3.78 [95 % CI 3.75, 3.83] for *Human* and 3.12 [95 % CI 3.07, 3.15] for *AI*. The rating distributions of the two main groups are shown in Figs. 2. Overall, *AI* ratings tended toward a normal distribution, while *Human* ratings were skewed more towards the right with a strong mode of 4 (see Fig. 2b).

3.3. Question-level analysis

The above-mentioned rating trends in the main groups were also observed, to a lesser degree at the levels of individual question, and case (see Fig. 3).

The per-question analysis showed significant rating differences ($p < 0.0033$) in favor of *Human* in all but three questions. These three exceptions were "relevant recommendations for further non-psychiatric treatment" (Q8, $p = 0.029$), "low proportion of unnecessary information" (Q13, $p = 0.49$) and "mentioning of relevant safety and legal aspects" (Q10, $p = 0.69$).

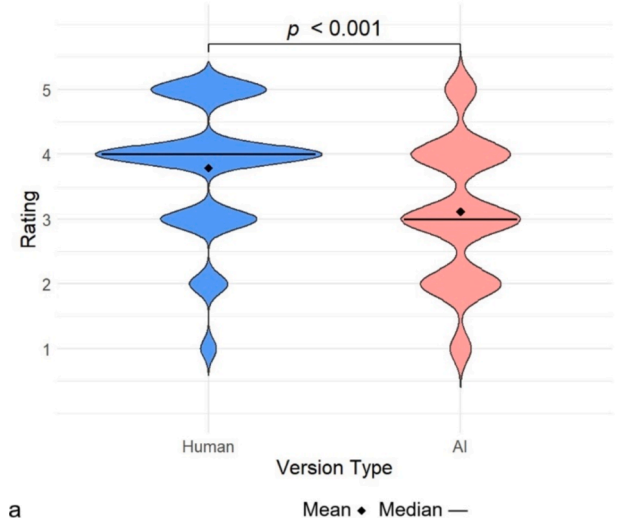


Fig. 2a. Violin plot of overall DS ratings with means and medians.

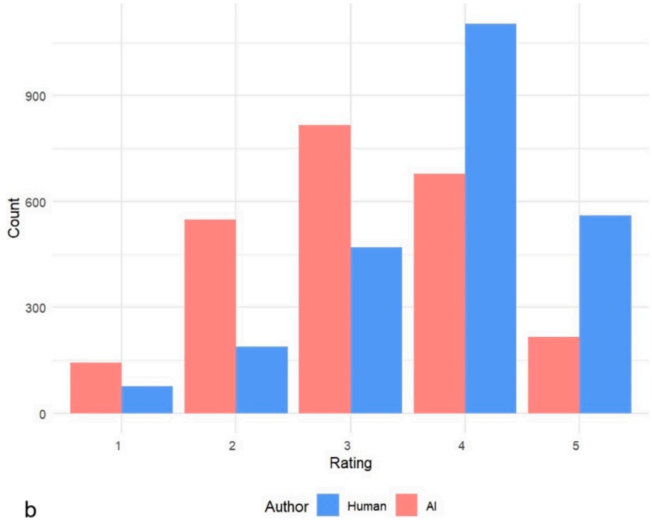


Fig. 2b. Bar chart of *Human* vs. *AI* DS rating distributions.

All other questions favored *Human* (see Table 4). The largest mean rating difference was observed in "specificity of formulations" (Q12, Δ1.19). The second and fourth largest differences occurred in the two prespecified questions most indicative of overall DS quality, "DS ready to send" (Q15, Δ1.12), and "low expected correction effort" (Q14, Δ1.05).

For "DS ready to send", when combining ratings ≥ 4 as "favorable" and ratings ≤ 2 as "unfavorable", overall approval ratings for *Human* DS were favorable at 61 %, neutral at 21 %, and unfavorable at 18 %. In contrast, for *AI*, the proportions were almost reversed (17 % favorable, 27 % neutral, 56 % unfavorable). For "low expected correction effort", approval ratings were slightly higher in both *Human* and *AI* (*Human* 67 % favorable, 19 % neutral, 14 % unfavorable, vs. *AI* 22 % favorable, 33 % neutral, 45 % unfavorable). The rating distributions of these two questions are shown in Fig. 4.

The other three highest mean rating differences, around Δ1, were observed in "quality of the summarization segment" (Q6, Δ1.07), "text coherence" (Q11, Δ0.96) and "mentioning of relevant instrumental and laboratory findings" (Q11, Δ0.94).

3.4. Case-level analysis

The per-case analysis showed significant rating differences in favor of *Human* in 16/20 cases ($p < 0.0025$). Three cases had no significant rating differences, and one case (C16) had significantly higher ratings for *AI* (see Table 5).

3.5. Rater-level analysis

In the per-rater analysis, all raters, except for one (with $p = 0.062$), rated *Human* DS significantly higher than *AI* with $p < 0.00625$.

The rating distributions per rater are depicted in the heatmaps in Fig. 5.

Raters' mean ratings are shown in the Supplementary Table 4. Supplementary Figure 2 provides an overview of all secondary outcomes with graphs correlating the mean ratings of *Human* with *AI* per question, case, and rater.

3.6. Qualitative review

In the unblinded qualitative case-based review, we identified hallucinations in 8/20 (40 %) of *AI*-DS and none in *Human*-DS. Of these hallucinations, 5 (62.5 %) were determined low clinical relevance, while 3 (37.5 %) were determined high clinical relevance. Additionally, we identified minor content errors in 6 (30 %) *AI*-DS and 2 (10 %) *Human*-DS.

4. Discussion

For the primary outcome, overall ratings favored *Human* significantly (mean_{Human} 3.78 vs. mean_{AI} 3.12), with the key threshold of 3 on our Likert scale marking the transition from neutral to mostly positive ratings. For the secondary outcomes, ratings also favored *Human* significantly in 12/15 questions. The greatest rating differences were observed in "specificity of formulations" (Δ1.19), "quality of the summarization segment" (Δ1.07), "text coherence" (Δ0.96), and the two prespecified questions most indicative of overall DS quality: "DS ready to send" (Δ1.12) and "low expected correction effort" (Δ1.05). There were no significant relationships between the two most common diagnoses, F20.0 and F31.2, their EHR lengths, or their ratings, nor was there a significant dependency between the EHR lengths of all individual cases and their ratings. However, this should be interpreted with caution due to the small sample size and our narrow input token range.

Overall, *AI* succeeded in formulating concisely, as evidenced by the item "the text contains little unnecessary information" (Δ0.07). It also performed reasonably well in sub-categories related to "pure

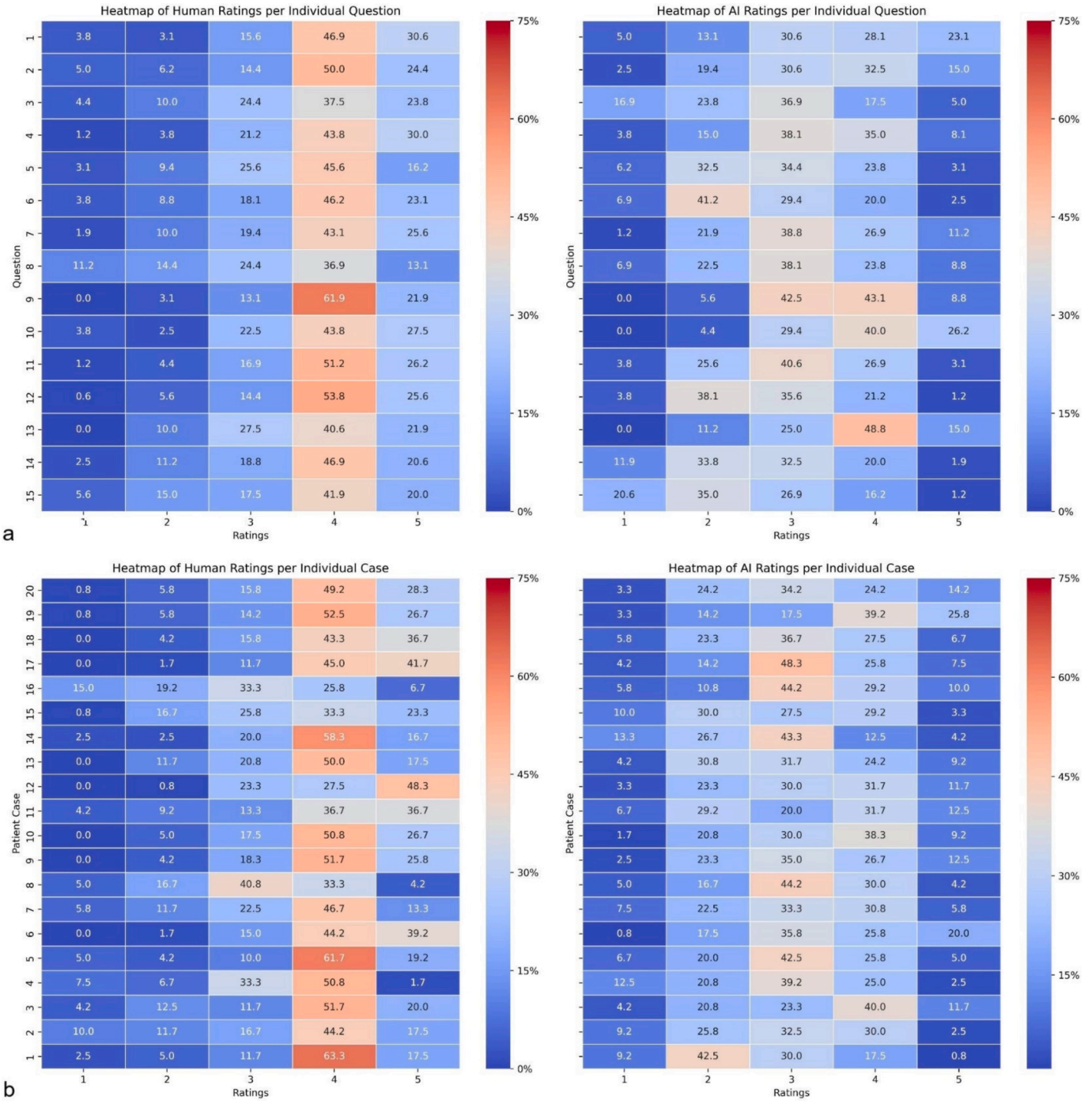


Fig. 3. Comparison of *Human* ratings (left) and *AI* ratings (right). The top two heatmaps (3a) show the rating distributions per individual question 1–15. The bottom two heatmaps (3b) show the rating distributions per individual case 1–20. Ratings are presented as percentages in relation to total ratings, with the scale maximum set at 75% to enhance the visibility of distributions.

information aggregation" and formalities, as seen in the items "assessment of clinical features" (Δ0.44), "assigning information to the correct diagnosis" (Δ0.45), "appropriate use of medical terminology" (Δ0.48), and "mentioning of relevant safety and legal aspects" (Δ0.01). The quality ratings dropped noticeably when this information then needed to be contextualized correctly, especially as EHR formulations increased in complexity. This aligns with recent findings indicating that the good performance of LLMs in classic reasoning tasks seems to be largely reliant on reproducing information from pretraining rather than on logical deduction [32,33]. Another clear shortcoming were the clinically relevant hallucinations that occurred in an average of 40 % of *AI*-DS (none in *Human*), of which 37.5 % were judged as "highly relevant". Similar to the study by Williams et al. [18], most hallucinations occurred in the final "recommendations section". Adhering to the prompt likely became more difficult toward the end of the LLM's response as it neared its context size maximum, a known issue in long-form generation tasks [33,34].

16/20 cases favored *Human* significantly. The unblinded EHR review revealed that in the one case in which AI performed better (Δ-0.44), the treating resident had changed shortly before the DS was written, giving the new physician little time to familiarize themselves with the patient, which might explain the disparity in performance. The other three cases

Table 4
Mean ratings with SD and mean rating differences $\Delta_{(Human\text{-}AI)}$ per individual question Q1-15 with the respective ICCs for *Human* and *AI*. Results are sorted from lowest to highest p values of the Wilcoxon signed-rank tests per question. ICCs < 0.5 are considered low, 0.5–0.75 moderate, and > 0.75 good rater agreement.

Questions	Mean Ratings *Human* ± SD	Mean Ratings *AI* ± SD	ICC *Human*	ICC *AI*	Mean Rating Difference $\Delta_{(Human\text{-}AI)}$	*p* values
Q12: Information is formulated specifically (and not too generally).	3.98 ± 0.83	2.79 ± 0.86	0.66	0.41	1.19	<0.0033
Q15: I would send out this discharge summary in its current form.	3.55 ± 1.14	2.43 ± 1.03	0.52	0.3	1.12	<0.0033
Q6: The summary gives me a good overview of the most important information of this case.	3.77 ± 1.02	2.70 ± 0.95	0.56	0.17	1.07	<0.0033
Q14: As a senior doctor, I would only have to make few corrections.	3.72 ± 1.00	2.67 ± 0.99	0.60	0.26	1.05	<0.0033
Q11: The text is coherent.	3.97 ± 0.84	3.01 ± 0.90	0.48	0.40	0.96	<0.0033
Q3: The relevant instrumental and laboratory findings are mentioned.	3.65 ± 1.08	2.71 ± 1.10	0.86	0.53	0.94	<0.0033
Q5: The relevant therapeutic effects and side effects are mentioned.	3.62 ± 0.97	2.85 ± 0.96	0.63	0.39	0.77	<0.0033
Q4: The relevant therapies are mentioned.	3.97 ± 0.88	3.29 ± 0.95	0.59	0.31	0.68	<0.0033
Q7: The relevant recommendations for further psychiatric treatment are mentioned, if possible, with name of outpatient practitioner and an appointment.	3.80 ± 0.99	3.25 ± 0.97	0.8	0.63	0.55	<0.0033
Q9: The text contains an appropriate amount of professional medical terminology.	4.03 ± 0.69	3.55 ± 0.73	0.49	0.33	0.48	<0.0033
Q1: The text describing each diagnosis is assigned correctly (it does not fit better to a different diagnosis).	3.97 ± 0.96	3.52 ± 1.13	0.63	0.59	0.45	<0.0033
Q2: The relevant clinical findings are mentioned.	3.82 ± 1.03	3.38 ± 1.04	0.74	0.44	0.44	<0.0033
Q8: The relevant recommendations for further non-psychiatric treatment are mentioned, if possible, with name of outpatient practitioner and an appointment.	3.26 ± 1.19	3.06 ± 1.04	0.82	0.63	0.2	0.029
Q13: The text contains little unnecessary information.	3.74 ± 0.91	3.67 ± 0.87	0.55	−0.02	0.07	0.49
Q10: Relevant safety concerns (endangering oneself or others) and other legally relevant information are mentioned.	3.88 ± 0.97	3.87 ± 0.85	0.63	0.05	0.01	0.69

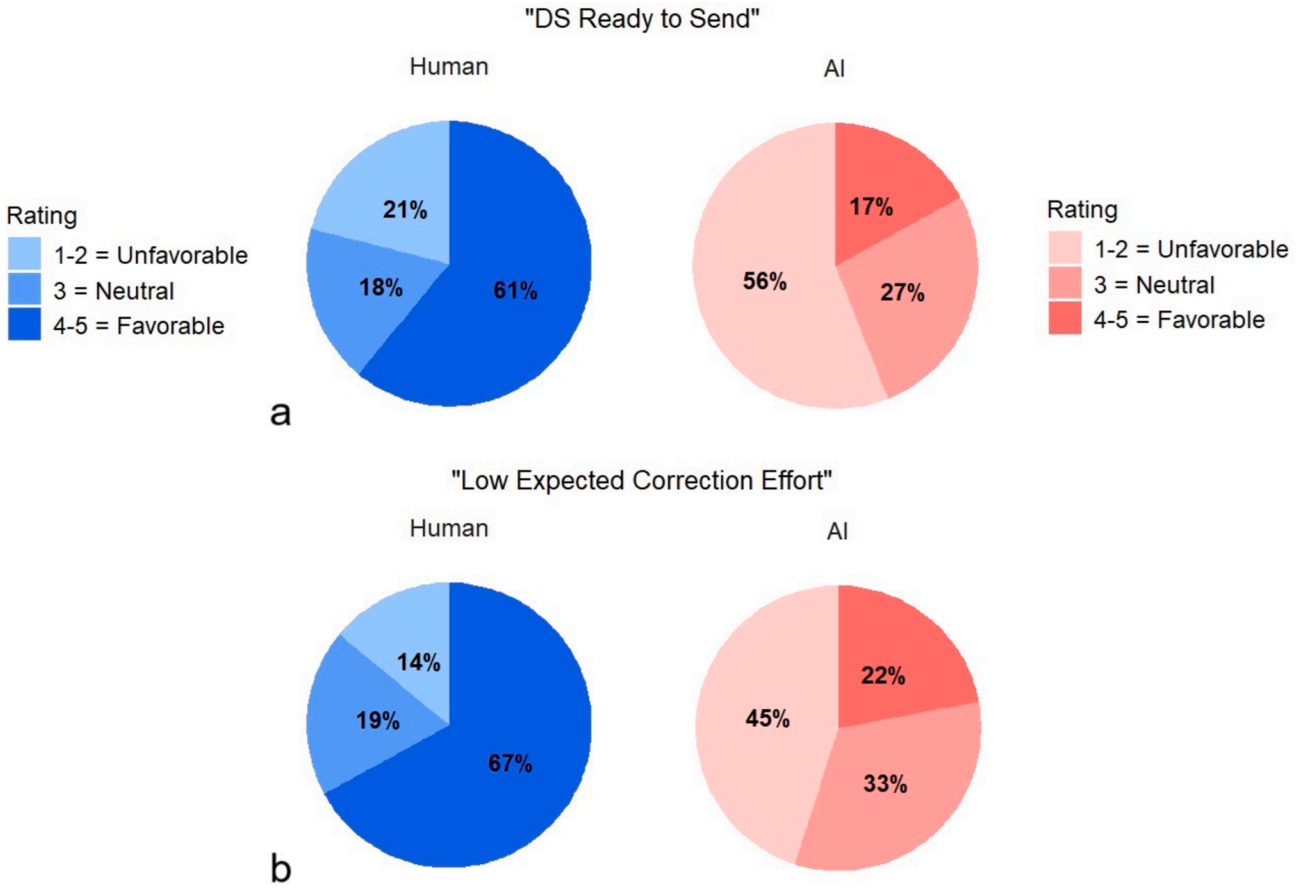


Fig. 4. Comparison of ratings for *Human* (left) and *AI* (right). The two upper pie charts (4a) show the distribution of ratings for "DS Ready to Send," while the two lower pie charts (4b) show the distribution for "Low Expected Correction Effort." Ratings are categorized as unfavorable (merge of ratings 1–2), neutral (rating 3), and favorable (merge of ratings 4–5).

Table 5
Mean ratings with SD and mean rating differences $\Delta_{(Human\text{-}AI)}$ per individual Case 1–20, sorted from lowest to highest p values of the respective Wilcoxon signed-rank test.

Case	Mean Ratings *Human* ± SD	Mean Ratings *AI* ± SD	Mean Rating Difference $\Delta_{(Human\text{-}AI)}$	*p* value
1	3.88 ± 0.94	2.58 ± 0.97	1.30	<0.0025
14	3.84 ± 0.95	2.68 ± 1.08	1.17	<0.0025
17	4.27 ± 0.95	3.18 ± 0.94	1.08	<0.0025
18	4.12 ± 0.99	3.06 ± 1.03	1.07	<0.0025
12	4.23 ± 1.02	3.25 ± 1.06	0.98	<0.0025
9	3.99 ± 0.97	3.23 ± 1.03	0.76	<0.0025
5	3.86 ± 1.05	3.02 ± 0.98	0.83	<0.0025
6	4.21 ± 0.98	3.47 ± 1.03	0.74	<0.0025
20	3.98 ± 1.02	3.22 ± 1.09	0.77	<0.0025
10	3.99 ± 0.98	3.32 ± 0.97	0.67	<0.0025
15	3.62 ± 1.12	2.86 ± 1.09	0.76	<0.0025
13	3.73 ± 0.98	3.03 ± 1.08	0.70	<0.0025
11	3.92 ± 1.22	3.14 ± 1.18	0.78	<0.0025
2	3.48 ± 1.26	2.91 ± 1.07	0.57	<0.0025
4	3,32 ± 0.98	2.84 ± 1.07	0.48	<0.0025
7	3.50 ± 1.13	3.05 ± 1.06	0.45	<0.0025
16	2.90 ± 1.15	3.27 ± 0.98	−0.37	<0.0025
19	3,98 ± 0.97	3.70 ± 1.12	0.28	0.0081
3	3.71 ± 1.16	3.34 ± 1.08	0.37	0.0158
8	3.15 ± 0.97	3.12 ± 0.94	0.03	0.6970

with non-significant rating differences in favor of *Human* (Δ0.28) showed very detailed and explicit EHR notes, which presumably facilitated DS generation for ChatGPT-4.

Our study had several limitations, including the small sample size, the token restriction, the low to moderate inter-rater agreement, the use of a non-specialized LLM, and no formal evaluation of information-completeness due to the blinding of raters. However, regarding the last point, residents base their DS not solely on EHR notes, which already impedes direct comparison. Another limitation were the comparatively low approval ratings for "DS ready to send" in *Human* (61 % favorable, 21 % unfavorable). For *AI*, proportions were almost reversed though (17 % favorable, 56 % unfavorable). The low *Human* approval ratings may be partially explained by the diverse psychiatric subspecialties of the raters, ranging from acute inpatient closed ward to outpatient treatment, and thus different DS expectations. On the other hand, having blinded raters without in-depth case knowledge, beyond the discharge letters' "clinical assessment" section, presumably improved rater objectivity.

Other strengths of our study included the use of real-life pseudonymized EHR data, the custom prompt, and the detailed Likert scale, which allowed for granular insight into the specific strengths and weaknesses of both groups, including ratings for potential time savings during correction. Furthermore, physicians did not know beforehand that their EHR notes would be used as a basis for DS, so our findings likely underestimate the potential of LLM-generated DS. For the future, results may improve if EHR notes were written with the specific goal of generating a DS in mind. It also seems expedient to develop custom LLMs specializing in medical DS generation. For instance, an open-source LLM could be fine-tuned on EHR-DS-pairs to specify the use case and to align DS outputs with a clinic's data and standards. This would also address data privacy problems, and pseudonymization might become unnecessary if computation was kept local. Additionally, results could be improved by incorporating different data than EHRs, such as instrumental diagnostics (high rating difference of Δ0.94 in this study) and the patient chart. In our internal preliminary tests, ChatGPT-4 was already able to correctly extract information about medication changes and vital parameters from partial patient chart screenshots and some newer LLMs like ChatGPT-4o or the announced "Project Astra" by Google process text, photo, and speech information multimodally within one model [35,36].

5. Conclusion

Overall, *AI*-DS did not achieve quality levels equivalent to those of psychiatric residents but performed comparably in 20 % of individual cases. Our findings of 22 % favorable and 33 % neutral ratings for "low correction effort" for *AI*-DS demonstrate potential for *AI*-DS to serve as DS-templates, which physicians could finalize, saving time in the future. Our results underscore the need for critical evaluation of LLM-outputs, ideally with in-depth case knowledge, as hallucinations and context errors are still frequent. The necessity for vigilance is highlighted by the

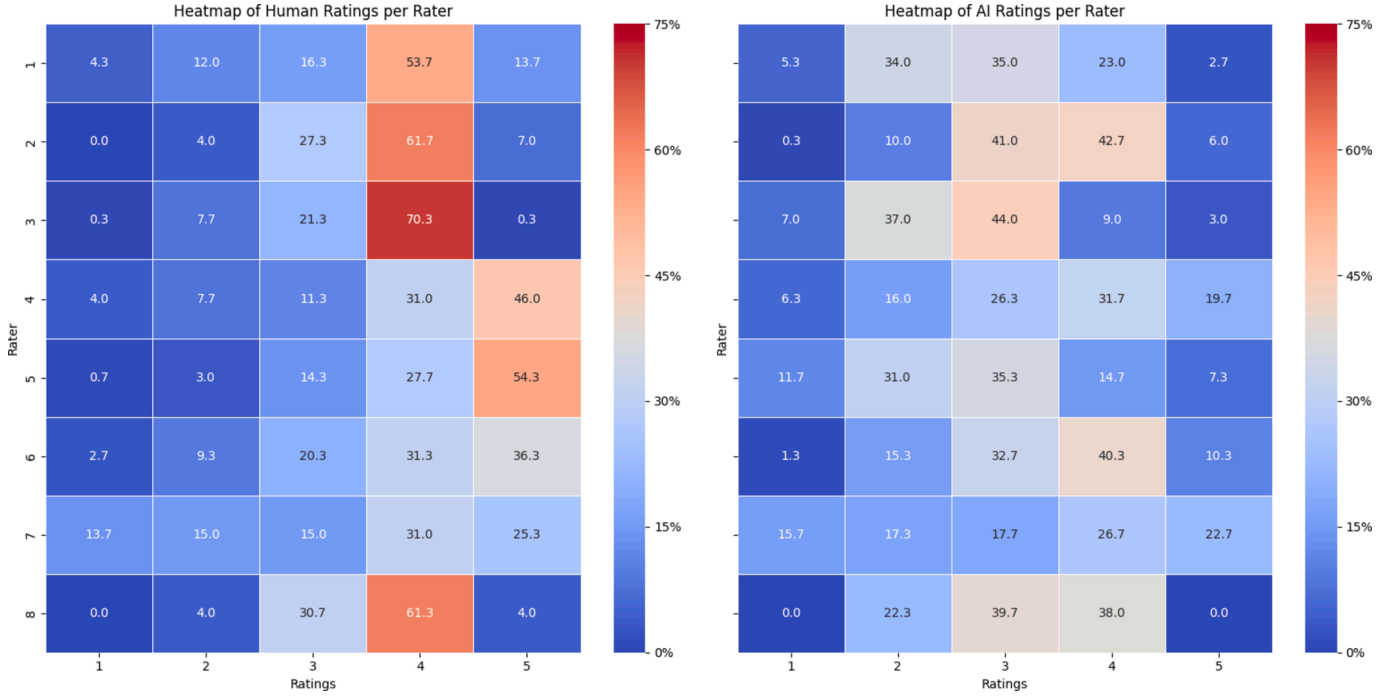


Fig. 5. Comparison of *Human* Ratings (left) and *AI* Ratings (right). The two heatmaps show the rating distributions per individual rater 1–8. Ratings are presented as percentages in relation to total ratings, with the scale maximum set at 75% to enhance the visibility of distributions.

fact that it is becoming increasingly difficult to distinguish LLM from human outputs. In our study, blinded raters only correctly identified *AI*-DS with a sensitivity of 81 % and a specificity of 75 %. Future studies should optimize the information basis, specialize the LLM for the use case, ensure assessment of information-completeness and hallucinations, and measure time savings to better harness the potential of LLMs in the clinical setting.

What was already known on the topic?

- DS contribute considerably to physicians' documentation burden
- Standardization can improve DS quality
- AI-DS frequently exhibit hallucinations and may omit clinically relevant information

What this study added to our knowledge

- Detailed and explicit formulations in EHRs can improve performance of AI-DS (20 % of cases with comparable ratings as residents)
- Strengths of AI-DS were information aggregation, conciseness, formalities, and structure
- Weaknesses of AI-DS were specificity, cohesion, and distillation of key information (summarization segment)
- With 22 % of AI-DS rated as "low expected correction effort," they show promise as DS-templates in a psychiatric setting

Declarations

Funding

No funding was received to assist with the preparation of this manuscript.

Data

The pseudonymized dataset of ratings and all codes for statistics can be made available upon request (e.g. Zenodo).

Ethics approval

Ethical approval was granted by the ethics committee of the Canton of Zurich (BASEC-ID 2024-00371) in view of the retrospective nature of the study.

Use of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used ChatGPT-4o to improve readability and language. The author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

CRediT authorship contribution statement

Arne Schwieger: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Katrin Angst:** Investigation, Conceptualization. **Mateo de Bardeci:** Writing – review & editing, Visualization, Validation, Methodology, Data curation, Conceptualization. **Achim Burrer:** Writing – review & editing, Resources, Investigation. **Flurin Cathomas:** Investigation. **Stefano Ferrea:** Investigation. **Franziska Grätz:** Investigation. **Marius Knorr:** Investigation. **Golo Kronenberg:** Writing – review & editing, Visualization, Validation. **Tobias Spiller:** Writing – review & editing, Visualization, Methodology, Investigation, Conceptualization. **David Troi:** Investigation. **Erich Seifritz:** Resources. **Samantha Weber:** Writing – review & editing, Visualization, Methodology. **Sebastian Olbrich:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijmedinf.2024.105654.

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